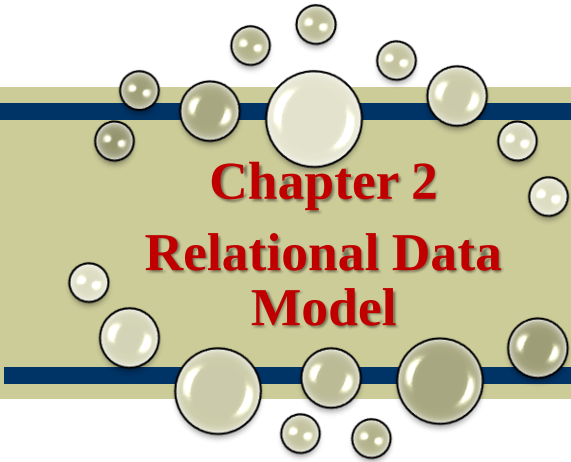




Chapter 2

Relational Data Model

2.8 TCL Commands: COMMIT, SAVEPOINT, ROLLBACK

2.8 DCL Commands: GRANT, REVOKE

2.9 DQL Commands: SELECT



Prepared By

Mohammad Ali

Lecturer (Selection Grade)

M. H. Saboo Siddik Polytechnic, Mumbai

Database Transactions

- A database transaction consists of one of the following:
 - DML statements that constitute one consistent change to the data
 - One DDL statement
 - One data control language (DCL) statement
- ◆ Begin when the first DML SQL statement is executed
- ◆ End with one of the following events:
 - A COMMIT or ROLLBACK statement is issued.
 - A DDL or DCL statement executes (automatic commit).
 - The user exits SQL.
 - The system crashes.

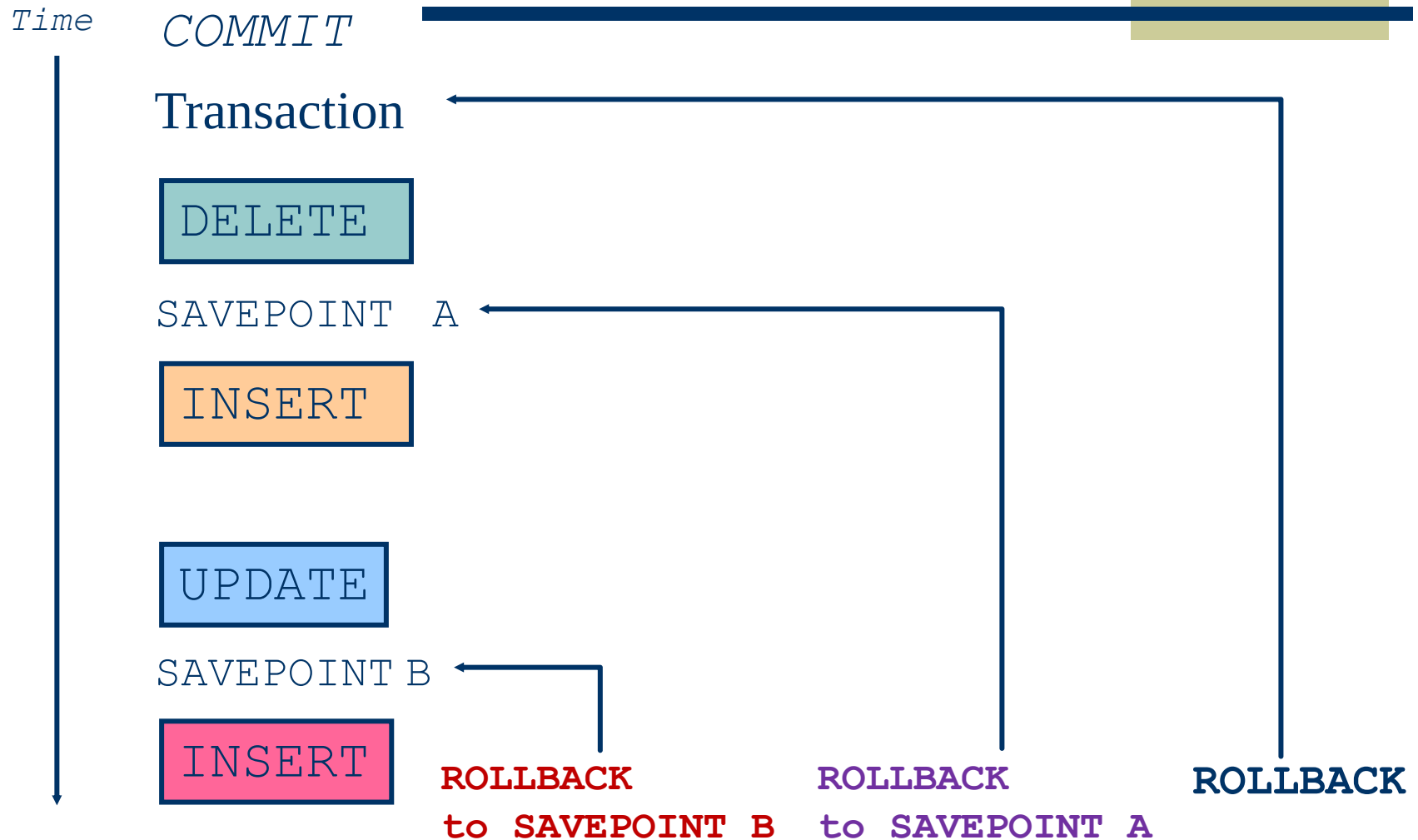
COMMIT & ROLLBACK Command

- INSERT, UPDATE, and DELETE commands are not committed to the database until the COMMIT statement is executed.
- COMMIT is a transaction managing command that confirms operations to the database on the server (closing Oracle also acts as a confirmation of the commands entered).
- SQL command ROLLBACK (ROLL) can be issued immediately to cancel any database operations since the most recent COMMIT.
- Like COMMIT, ROLLBACK is also a transaction managing command; however, it cancels operations instead of confirming them.

Advantages of COMMIT and ROLLBACK Statements

- ◆ With COMMIT and ROLLBACK statements, we can:
 - Ensure data consistency.
 - We have control over making changes to the data permanent
 - We can preview data changes before making it permanent
 - Group logically related operations

Controlling Transactions



Rolling Back Changes to a Marker

- ◆ Create a marker in a current transaction by using the **SAVEPOINT** statement, which divides the transaction into smaller sections.
- ◆ Discard pending changes to that marker by using the ROLLBACK TO SAVEPOINT statement.
- ◆ A second savepoint with the same name as an earlier savepoint, the earlier savepoint is deleted.

```
UPDATE...
```

```
SAVEPOINT update_done;
```

```
Savepoint created.
```

```
INSERT...
```

```
ROLLBACK TO update_done;
```

```
Rollback complete.
```

Implicit Transaction Processing

- An automatic commit occurs under the following circumstances:
 - DDL statement is issued
 - DCL statement is issued
 - Normal exit from SQL, without explicitly issuing COMMIT or ROLLBACK statements
- An automatic rollback occurs under an abnormal termination of SQL or a system failure.

State of the Data Before COMMIT or ROLLBACK

- The previous state of the data can be recovered.
- The current user can review the results of the DML operations by using the `SELECT` statement.
- Other users *cannot* view the results of the DML statements by the current user.
- The affected rows are *locked*; other users cannot change the data in the affected rows.

State of the Data After COMMIT

- Data changes are made permanent in the database.
- The previous state of the data is permanently lost.
- All users can view the results.
- Locks on the affected rows are released; those rows are available for other users to manipulate.
- All savepoints are erased.

Committing Data

♦ Make the changes:

```
DELETE FROM employees
WHERE employee_id = 99999;
1 row deleted.
```

```
INSERT INTO departments
VALUES (290, 'Corporate Tax', NULL, 1700);
1 row created.
```

```
DELETE FROM departments
WHERE department_id IN (290, 300);
1 row deleted.
```

```
UPDATE employees
SET department_id = 80
WHERE employee_id = 206;
1 row updated.
```

♦ Make the data changes permanent:

```
COMMIT;
```

```
Commit complete.
```

State of the Data After ROLLBACK

- Discard all pending changes by using the **ROLLBACK** statement:
 - Data changes are undone.
 - Previous state of the data is restored.
 - Locks on the affected rows are released.

```
DELETE FROM copy_emp;
```

```
22 rows deleted.
```

```
ROLLBACK ;
```

```
Rollback complete.
```

State of the Data After ROLLBACK

```
DELETE FROM test;  
25,000 rows deleted.
```

```
ROLLBACK;  
Rollback complete.
```

```
DELETE FROM test WHERE id = 100;  
1 row deleted.
```

```
SELECT * FROM test WHERE id = 100;  
No rows selected.
```

```
COMMIT;  
Commit complete.
```

While attempting to remove a record from the TEST table, we can accidentally empty the table. We can correct the mistake, reissue the proper statement, and make the data change permanent.

Statement-Level Rollback

- ◆ If a single DML statement fails during execution of a transaction, only that statement is rolled back by statement-level rollback.
- ◆ The Oracle server implements an implicit savepoint.
- ◆ Part of a transaction can be discarded by an implicit rollback if a statement execution error is detected
- ◆ All other changes are retained.
- ◆ The user should terminate transactions explicitly by executing a COMMIT or ROLLBACK statement.

DCL: Data Control Language – GRANT & REVOKE

- ◆ Controlling Access to database objects such as tables and views
- ◆ Example : Granting “SCOTT” the access to Table “student” (for inserting, updating and deleting)
 - GRANT INSERT, UPDATE, DELETE ON Student TO SCOTT
 - GRANT <privileges> ON <object name>
TO <grantee> [<comma> <grantee> ...]
[WITH GRANT OPTION]
 - WITH GRANT OPTION: allows the grantee to further grant privileges
 - Can be limited to a column of a table, Ex: GRANT UPDATE(name) ON Student TO Mary
 - To revoke privileges : REVOKE